

Generation Interconnection Request System Impact Study Report

For Queue Project Q13-0212

Sabanera 230 kV
August 30, 2014

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q13-0212. The Interconnection Customer proposes a 47.7 MW solar photovoltaic facility interconnecting at the Sabanera 230 kV substation in the Indiana area. The request was received on March 11, 2013. The study evaluated system intact and single contingency conditions using the regional planning case, including 150 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 17 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$19,511,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

The LARSON STREET to NORWICH 230 KV is overloaded to 115.0% of its 738 MVA emergency rating for the loss of the NORWICH SUBSTATION to BARRETT 500 KV Line. The project contributes 11.7% DFAX (5.6 MW) to this overload.

For the loss of BARRETT to NORWICH 500 KV, the MINCO to LARSON ST 230 KV L loads to 112.3% of its 786 MVA rating. Project DFAX: 7.8%.

For the loss of BEAVER to TREGO 500 KV, the MYRTLE POINT to GALION GENERATING STA 230 KV loads to 108.9% of its 628 MVA rating. Project DFAX: 6.4%.

The MADELINE to E METHUEN 230 KV is overloaded to 108.5% of its 607 MVA emergency rating for the loss of the EAST METHUEN to GRAY'S FLAT 230 KV. The project contributes 15.0% DFAX (7.2 MW) to this overload.

For the loss of PINEY CRK PROJECT to HILLTOP 500 KV, the PINEY CREEK PROJECT 500/345kV Autotransformer loads to 107.1% of its 1,257 MVA rating. Project DFAX: 9.7%.

For the loss of IMPERIAL to EUFAULA 500 KV, the EUFAULA 230/1kV Autotransformer loads to 106.3% of its 1,126 MVA rating. Project DFAX: 11.6%.

For the loss of CARBON CENTER 500 kV/345 kV TX, the BLACKROCK to SHAWVILLE STA 345 KV loads to 104.8% of its 1,099 MVA rating. Project DFAX: 8.4%.

For the loss of FT HUMBUG to ORCHARD ROAD SUBSTATION 500 KV L, the FORT HUMBUG 500-345 kV Transformer loads to 104.1% of its 1,123 MVA rating. Project DFAX: 5.2%.

The SHAWVILLE STA 500/345 kV Auto is overloaded to 103.3% of its 724 MVA emergency rating for the loss of the CARBON CENTER 500 kV/345 kV Transformer. The project contributes 5.4% DFAX (2.6 MW) to this overload.

KAPSVILLE to MORENO TAP 230 KV Line: 102.3% loading (570 MVA) following the outage of the TREGO to SWAN VALLEY 500 KV; DFAX = 8.5%.

For the loss of SWAN VALLEY to TREGO 500 KV, the EASTOWN ROAD to MORENO TAP 230 KV loads to 102.3% of its 592 MVA rating. Project DFAX: 8.5%.

TALENENERG MARTINS CREEK LLC WILLIAMSPORT to MONTOUR STATION 345 KV: 101.9% loading (1,130 MVA) following the outage of the ORCHARD RD to FORT HUMBUG 500 KV; DFAX = 5.9%.

KAMPSVILLE to ZONA LIBRE 6007 230 KV L: 101.7% loading (562 MVA) following the outage of the SWAN VLY to TREGO 500 KV line; DFAX = 8.5%.

For the loss of MOUNTAN to FORT HUMBUG 500 KV line, the LITTLE CHUTE to FT HUMBUG 345 KV line loads to 101.3% of its 1,440 MVA rating. Project DFAX: 5.3%.

For the loss of SWAN VLY to TREGO 500 KV, the GEORGE NEAL S to ZONA LIBRE 6007 230 KV loads to 101.2% of its 543 MVA rating. Project DFAX: 8.5%.

STRAWBERRY BELT to HILLTOP 345 KV line: 100.3% loading (1,135 MVA) following the outage of the HILLTOP to TALENENERGY MARTINS CREEK LLC WILLIAMSPORT 500 KV; DFAX = 6.5%.

For the loss of SWAN VALLEY to TREGO 500 KV, the MYRTLE POINT to GEORGE NEAL S 230 KV loads to 100.0% of its 528 MVA rating. Project DFAX: 8.5%.

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
LARSON STREET to NORWICH 230 KV	Rebuild line with higher-capacity conductor	\$11,346,000	2% (\$170,000)
MINCO SUBSTATION to LARSON STREET 230 KV	Rebuild line with higher-capacity conductor	\$10,288,000	2% (\$208,000)
GALION GENERATING STA to MYRTLE POINT 230 KV Line	Reconductor line	\$6,890,000	2% (\$106,000)
MADELINE to E METHUEN 230 KV	Reconductor line	\$9,954,000	2% (\$169,000)
PINEY CREEK PROJECT 500-345 kV Transformer Bank	Replace transformer with higher-rated unit	\$29,452,000	2% (\$437,000)
EUFAULA SUBSTATION 230-1 kV Transformer Bank	Replace transformer with higher-rated unit	\$8,051,000	2% (\$118,000)
BLACKROCK to SHAWVILLE STA 345 KV	Reconductor line	\$14,034,000	2% (\$210,000)
FORT HUMBUG 500-345 kV Transformer	Replace transformer with higher-rated unit	\$19,208,000	16% (\$2,994,000)
SHAWVILLE STA 500/345kV Transformer Bank	Replace transformer with higher-rated unit	\$46,069,000	16% (\$7,151,000)
MORENO TAP to KAMPSVILLE 230 KV line	Reconductor line	\$13,196,000	2% (\$262,000)
EASTOWN ROAD to MORENO TAP 230 KV	Reconductor line	\$26,525,000	2% (\$526,000)
TALENENERG MARTINS CREEK LLC WILLIAMSPORT to MONTOUR STATION 345 KV	Reconductor line	\$28,584,000	2% (\$430,000)
KAMPSVILLE to ZONA LIBRE 6007 230 KV L	Reconductor line	\$9,224,000	2% (\$183,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
LITTLE CHUTE to FT HUMBUG 345 KV line	Reconductor line	\$38,574,000	16% (\$5,998,000)
GEORGE NEAL S to ZONA LIBRE 6007 230 KV	Reconductor line	\$14,183,000	2% (\$281,000)
STRAWBERRY BELT SUBSTATION to HILLTOP 345 KV	Reconductor line	\$11,265,000	2% (\$169,000)
MYRTLE POINT to GEORGE NEAL S 230 KV	Reconductor line	\$4,993,000	2% (\$99,000)
	Total allocated to project		\$19,511,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.